10.Grade the pitting edema according to the following scale: - 1+ = Minimal edema of the lower extremities (at pedal and pretibial sites). - 2+ = Marked edema of the lower extremities. - 3+= edema of the lower extremities, face, hands, and sacral area. - 4+ = Generalized, massive edema.				
Post Procedure Task				
13. Record and. compare the finding with those previously recorded.				
14. Cover the woman and keep her in comfortable position.				
15.Reassure mother about result.				
Total score:				
Date:				

Student Name **Instructor Name**

6} Deep Tendon Reflex Assessment

Goals:

Objectives:

- To assess CNS status in cases of PIH, hyperemiss gravidarum
- To assess CNS status in cases woman who take Mgso4 through (patellar) [knee jerk] and biceps reflexes).

Out lines:

- Introduction.
- Definition of reflex, deep tendon reflex. Importance of reflex in human body.
- Equipment used in the procedure.
- Procedure.
- Conclusion.

Introduction:

Hyperreflexia refers to hyperactive or repeating (clonic) reflexes. These usually indicate an interruption of corticospinal and other descending pathways that influence the reflex arc due to a suprasegmental lesion, that is, a lesion above the level of the spinal reflex pathways.

Definitions of:-

- Reflex:

Is involuntary and largely unconscious reaction in response to a peripheral stimulus; detected by an effector organ.

- Deep tendon reflex:

DTRs (myotatic or stretch reflexes)

- It is involuntary muscle contractions in response to sudden tendon stimulation induced by a sharp tap by a finger or rubber hammer on the tendon of insertion of the muscle.
- They help assess the integrity of the spinal cord, peripheral nerves, and the neuromuscular junction.

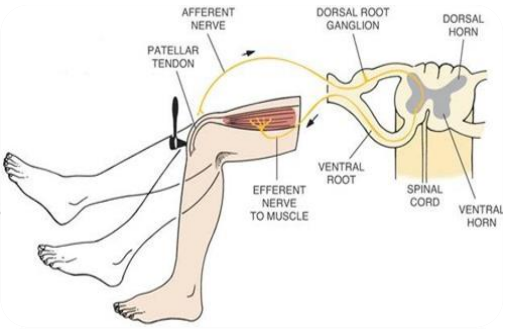
Sites:

- ✓ Upper Extremities:
  - Biceps (musculocutaneous nerve): Contraction of the biceps muscle causes elbow flexion.
  - Brachioradialis (radial nerve): Forearm flexion and slight supination.
  - Triceps (radial nerve): Contraction of the triceps leads to elbow extension.
- ✓ Lower Extremities:
  - Patella (femoral nerve): Knee extension in response to a tap on the patellar tendon.
  - Ankle (tibial nerve): Plantar flexion of the foot when the Achilles tendon is tapped.

By convention the deep tendon reflexes are graded as follows:

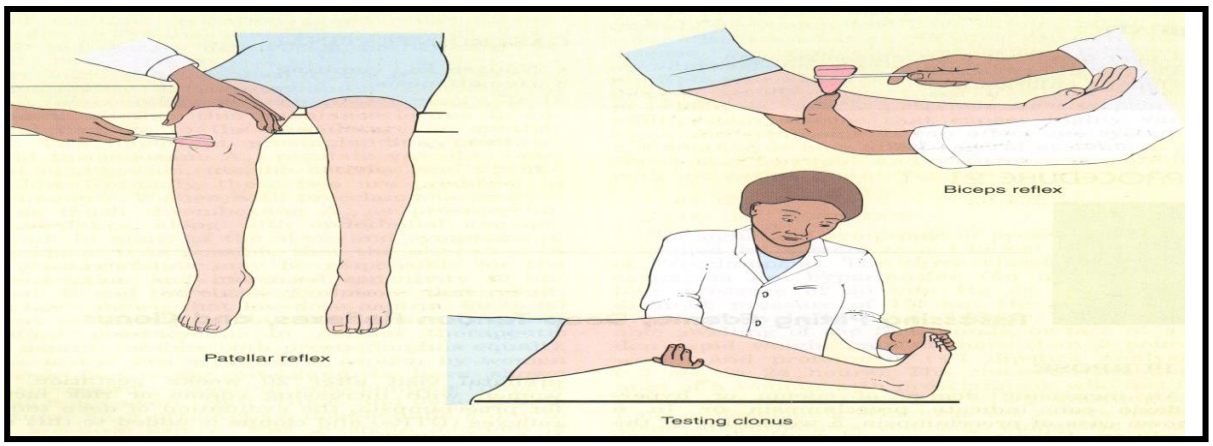
- 0 = no response; always abnormal
- 1+ = a slight but definitely present response; may or may not be normal
- 2+ = a brisk response; normal
- 3+ = a very brisk response; may or may not be normal
- 4+ = a tap elicits a repeating reflex (clonus); always abnormal

1. Rate the reflex with the following scale:
   - 5+ Sustained clonus



- 4+ Very brisk, hyper-reflexive, with clonus
- 3+ Brisker or more reflexive than normally.
- 2+ Normal(negative clonus)
- 1+ Low normal, diminished
- 0.5+ A reflex that is only elicited with reinforcement

No response



Importance of reflex in human body:

1) Used to assess nervous system status, and detect any abnormalities.

2) To detect MGSO4 toxicity.

➢ Assess the biceps reflex-

- The forearm should be resting on the client's trunk.
- Place the thumb firmly on the client's biceps tendon (antecubital space) and strike the thumbnail briskly with the reflex hammer.
- The biceps muscle will respond by slight movement. The lower arm should flex in response.

➢ Assess the knee jerk reflex or patellar

tendon reflex.

- The knee should be positioned halfway between the longest and shortest positions.
- Support is given under the knee with the foot off the bed (45-degree angle).
- The patellar tendon is struck (tapped) just below the patella, and the quadriceps muscle group should be observed for contraction (slight movement). The lower leg should extend in response.

➢ **Clonus is the sudden stretching of a hypertonic muscle, producing reflex contraction.**

If the stretch is maintained during subsequent relaxation, further reflex contraction occurs, and this may continue almost indefinitely, unless the stretch stimulus is released. It is demonstrated by dorsiflexion of the foot or by sharply moving the patella downward, but it may be present at any joint. Clonus represents an increase in reflex excitability and may be present in an extremely tense client.

Equipment:

1. Disposable gloves.
2. Percussion hammer.

5) Deep Tendon Reflex Assessment

Action	First Trial		Second Trial	
	Done	Not done	Done	Not done
Preparatory phase				
2. Prepare the necessary equipment (reflex hammer).				
3. Greet the mother.				
4. Explain the procedure to the mother and reassure her overall the procedure.				
5. keeping her privacy (close windows, doors, put curtain)				
6. Hand washing and keep it warm.				
7. Promote relaxation for proper testing, because these reflexes are difficult to elicit when the client is tense.				
1. Assess the biceps reflex- ☞ The forearm should be resting on the client's trunk. ☞ Place the thumb firmly on the client's biceps tendon (antecubital space) and strike the thumbnail briskly with the reflex hammer. ☞ The biceps muscle will respond				

by slight movement. ☞ The lower arm should flex in response.				
8. **Assess the knee jerk reflex or patellar tendon reflex.** ☞ The knee should be positioned halfway between the longest and shortest positions. ☞ Support is given under the knee with the foot of the bed (45-degree angle). ☞ The patellar tendon is struck (tapped) just below the patella, and the quadriceps muscle group should be observed for contraction (slight movement). The lower leg should extend in response.				

2. Clonus is the sudden stretching of a hypertonic muscle, producing reflex contraction. ☞ Support her leg by placing one hand or arm under her knee. ☞ Using other hand, sharply dorsiflex the woman's foot, maintaining slight pressure on her foot after dorsiflexion. ☞ Release her foot and observe its movement. ☞ Count the beats of Clonus present as observe for rhythmic contraction and relaxation of the gastrocnemius and soleus muscles.				
Post Procedure Task				
9. Determine the degree of DTR 10. Record and compare findings with those previously recorded. 11. Remove equipment. 12. Wash hand. 13. Reassure mother about findings.				
Total score:				
Date:				

Student Name **Instructor Name**